REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of different books, articles, and previous studies related to the proposed SMARTENROLL: Web-Based Enrollment Management System for Adreo Montessori Inc. The purpose of this chapter is to give a clear background of the study by discussing ideas and systems related to enrollment management, web-based systems, database management, cloud storage, data security, and educational information systems.

The literature reviewed in this chapter helps explain how technology can improve the enrollment process. It focuses on important concepts such as automation, centralized data storage, real-time monitoring, and secure handling of information. These ideas support the development of a system that is more accurate, efficient, and easier to use compared to manual enrollment processes.

This chapter also includes both foreign and local studies that show how different schools and institutions use web-based enrollment systems and other digital platforms. These studies help the researchers understand how similar systems work and how they can be improved or applied to the SMARTENROLL system.

The information gathered from these related literature and studies serves as a guide in designing and developing the proposed system. Through this, SMARTENROLL aims to provide a faster, more organized, and more reliable enrollment process while improving the overall experience of both administrators and users.

2.1 RELATED LITERATURES

Foreign Literature

In today's educational environment, enrollment management systems have become essential tools for improving the efficiency and organization of administrative processes in schools. Many institutions are now adopting web-based and data-driven systems to replace traditional manual enrollment procedures, allowing faster processing, better data accuracy, and easier access to student information. According to Smith and Brown (2022), modern enrollment management systems provide a centralized platform where student data can be collected, stored, and analyzed to support decision-making related to admissions, resource allocation, and student retention. Their study also emphasized that features such as real-time data access, automated validation, and report generation help reduce errors, minimize administrative workload, and improve overall system transparency. As a result, these systems enable educational institutions to respond more effectively to changes in student population and improve the quality of services provided to students and parents.

This foreign literature supports the development of the SMARTENROLL system by showing how web-based enrollment management systems can improve the overall efficiency and accuracy of school processes. By using a centralized database, real-time data access, and automated features such as validation and report generation, the system can reduce manual errors, minimize administrative workload, and make enrollment faster and more organized. These ideas are directly applicable to the project since SMARTENROLL aims to provide a more reliable, user-friendly, and efficient solution for managing student enrollment and records in Adreo Montessori Inc.

To complement this approach, Chamilco (2023) focused on the application of digital enrollment systems, which automate the traditional registration processes for students. According to the study, digital enrollment systems work in a systematic manner that involves gathering requirements, designing, and programming to enable students to register online and provide the necessary information. The significance of digital enrollment systems is that they eliminate mistakes, decrease queues, and make enrollment information more accurate and

accessible. Furthermore, digital enrollment systems are advantageous to institutions because they improve the efficiency, transparency, and convenience of the administrative processes of institutions. The two studies complement each other in that they demonstrate that contemporary enrollment management systems not only optimize processes but also enable strategic planning through data analysis and digital workflow. This is applicable to the development of an enrollment management system project because it ensures that the system is optimized through analysis and digital capabilities to process enrollments efficiently, maintain accurate records, and make informed decisions for institutions of higher learning.

This foreign literature supports the SMARTENROLL system by highlighting the importance of having a structured and well-designed enrollment process that follows clear steps such as data collection, system development, and implementation. It shows that using a digital approach can improve how information is organized and managed, making the enrollment process more consistent and reliable. For the project, this idea helps ensure that SMARTENROLL is not only efficient but also properly designed to handle student data in an organized way, making it easier for administrators to manage records and maintain smooth enrollment operations.

Local Literature

In the Philippines, the integration of digital technology in school administration has become more important to enhance efficiency and service delivery. Canizo et al., (2024) have conducted a study on the enrollment management system of Novaliches High School, emphasizing the drawbacks of manual and paper-based enrollment processes. The study clearly pointed out that manual processes entail long queues and waiting times, thus creating inefficiencies for students as well as administrative personnel. To overcome these issues, the researchers have developed a digital Enrollment Management System where students can enroll online, check their enrollment status, and provide required information online. The system has shown remarkable improvements in terms of processing time, accuracy, and overall efficiency. This study is very much relevant in the context of the Philippines, where manual enrollment systems are still being used in schools, which are time-consuming and error prone.

This local literature supports the SMARTENROLL system by showing how replacing manual, paper-based enrollment with a digital system can significantly improve efficiency and service delivery in schools. It highlights those common problems such as long queues, delays, and errors can be reduced through an organized and automated process. This is helpful to the project because SMARTENROLL also aims to address these same issues in Adreo Montessori Inc. by providing a faster, more accurate, and more convenient way of managing enrollment, making the overall process easier for both administrators and parents.

As well, Polinio et al., (2023) investigated the effects of digital transformation on the efficiency of administrative work in public schools in Bukidnon. The findings of the study showed that the application of digital technology for record-keeping, data management, and communication helped to improve the efficiency of the workflow, minimize manual work, and enabled the staff to manage administrative work more effectively. This study highlights the increasing trend of adopting digital solutions in Philippine educational institutions.

This literature is useful for the SMARTENROLL system because it demonstrates how digital tools can improve the efficiency of school administrative tasks by reducing manual work and streamlining workflow. By using technology for record-keeping, data management, and communication, school staff can manage information more accurately and save time. These findings support the development of SMARTENROLL, as the system aims to provide a digital platform for enrollment management that enables faster processing, organized records, and more effective administration in schools.

2.2 RELATED STUDIES

Foreign Studies

In other countries, the application of web-based systems in educational and organizational environments has been proven to enhance efficiency, accuracy, and service delivery. Syafariani and Devi (2019) carried out a study on a Web-Based Academic Information System at Universitas Komputer Indonesia, concentrating on the automation of academic processes such as class scheduling, grade management, and report generation. The system was developed using PHP and MySQL, enabling students and academic staff to manage academic information online. The findings indicated that the system was able to significantly decrease the manual workload, processing time, and human errors, resulting in faster and more accurate academic processes.

This study is relevant to the SMARTENROLL system because it demonstrates how web-based platforms can streamline academic and administrative processes by reducing manual work and errors. By automating tasks like data management, scheduling, and reporting, such systems improve efficiency, accuracy, and accessibility of information for both staff and students. These insights support the development of SMARTENROLL, as the system aims to provide a similar web-based solution for enrollment management, ensuring faster processing, accurate records, and more efficient handling of student information.

In addition, the effectiveness of web-based enrollment systems in enhancing student enrollment and administrative efficiency has been examined in several international settings. For instance, a study by Al-Badri et al., (2020) examined the effectiveness of a web-based student enrollment system intended to facilitate the online management of course enrollment, class schedules, and student information. The proposed system was expected to eliminate the need for manual processing, reduce errors, and optimize the student enrollment process for both students and administrative personnel. The findings of the study confirmed that the system was effective in reducing the time spent on registration, increasing the accuracy of student information, and optimizing administrative processes.

This foreign study is useful for the development of the SMARTENROLL system because it highlights how web-based enrollment platforms can improve both student and administrative experiences. By automating course registration, schedule management, and student information handling, such systems reduce errors, save time, and streamline administrative tasks. These findings support the design of SMARTENROLL, showing that a digital enrollment system can enhance efficiency, ensure accurate records, and make the enrollment process faster and more convenient for students and school staff alike.

Local Studies

Within the Philippine educational context, institutions have developed automated enrollment solutions to address the limitations of traditional manual processes. One such study is Enrollment Management System of Bestlink College of the Philippines for Senior High School, authored by Cavite et al., (2024). The study focused on designing and developing an enrollment management system that consolidates the entire enrollment process into a single platform, allowing students to complete registration tasks more efficiently. The system integrates online and offline procedures, analyzes student enrollment requirements, and undergoes testing prior to implementation to ensure optimal functionality. It also includes maintenance support to address operational issues. The significance of this system lies in its ability to streamline enrollment operations, reduce administrative workload, and minimize errors commonly associated with manual processes. The study demonstrates how a centralized digital enrollment platform can enhance operational efficiency and improve data management in academic institutions.

The findings of this study are valuable for the SMARTENROLL project because they highlight how a unified digital enrollment platform can make registration processes faster and more organized. By integrating both online and offline procedures, addressing student requirements, and providing system support, such platforms help reduce errors and lighten the administrative workload. These insights can guide the development of SMARTENROLL to ensure it efficiently manages student enrollment, maintains accurate records, and improves overall school operations.

Building on these developments, web-based enrollment systems have also been implemented in other Philippine educational institutions to further enhance administrative efficiency and data accuracy. Javelosa et al., (2018) developed a Web-Based Enrollment System with Student Information System (SIS) for the Children's Integrated School of Alta Tierra (CISAT). This system integrates key modules, including student profiling, registration, scheduling, updates, and payment management, within a web-based platform using ASP.NET and MySQL. It enables students to update their profiles online while allowing administrators to monitor academic requirements and schedules efficiently. The study highlights the importance of integrating enrollment functions with a student information system to facilitate real-time data management, reduce redundancy, and improve access to accurate student records. It further demonstrates how web-based systems effectively address the challenges of manual enrollment, ensuring faster, more accurate, and more organized school operations.

This local study is relevant to the SMARTENROLL project as it demonstrates how the integration of enrollment processes with a Student Information System can enhance the efficiency and accuracy of school operations. By enabling students to manage their profiles online and providing administrators with real-time access to academic records, such systems reduce manual workload, minimize data redundancy, and facilitate faster processing. These findings provide a valuable framework for the design and development of SMARTENROLL, supporting the creation of a centralized and user-friendly platform that streamlines enrollment procedures, ensures accurate data management, and promotes more effective administrative decision-making.